

Detailed Syllabus:**UNIT-I: Introduction to Software Engineering**

Need of software engineering, systems engineering, challenges in software engineering, Software process models, quality characteristics of software systems, Ethics in Software Engineering.

UNIT-II: Requirements Engineering

Requirements engineering process, requirements specification, structured and object oriented analysis

UNIT-III: Software Design

Architectural design, detailed design, Structured and object oriented design, user interface design

UNIT-IV: Software Testing

Verification, Validation, testing techniques, Testing Process

UNIT-V: Tools and Evolution

CASE Tools, Reverse engineering, Reengineering and Configuration management.

Reference Books:

1. Ian Sommerville (2016), "*Software Engineering*", 10th Edition, Pearson Education Limited, Global Edition.
2. Roger S Pressman, Bruce R Maxim(2015), "*Software Engineering, A Practitioner's Approach*", 8th Edition, TataMcGraw Hill, Indian Edition
3. Grady Booch, James Rumbaugh, Ivor Jacobson(2005), "*The Unified Modeling Language User Guide*", 2nd Edition, Addison Wesley Professional.US